

## MINERALOGY AND PETROLOGY OF SAMPLE 67075 AND THE ORIGIN OF LUNAR ANORTHOSITES

I.S. McCALLUM, F.P. OKAMURA and S. GHOSE

*Department of Geological Sciences, University of  
Washington, Seattle, Wash. (USA)*

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Plagioclase in cataclastic anorthosite 67075 occurs as angular matrix grains and as recrystallized clasts of micro-anorthosite. Olivines are Fe-rich and fall into two compositional groupings. Large grains of pyroxene show exceptionally well-developed exsolution lamellae analogous to those observed in pyroxenes from layered complexes. The low-Ca component in both pigeonites and augites shows varying degrees of inversion to orthopyroxene. The lattices of host and lamellae may deviate slightly (up to  $2^\circ$ ) from the ideal orientation. Very slow cooling from magmatic temperatures is required to produce the coarse exsolution textures and inversion features. Augite macrocrystals are distinctly subcalcic indicating crystallization at temperatures around  $1100 \pm 50^\circ\text{C}$  while host-lamellae pairs and small grains in lithic clasts and matrix indicate reequilibration on a micron scale to temperatures less than  $800^\circ\text{C}$ . Pyroxene compositions tend to cluster into two groups both of which are among the most Fe-rich reported for highland pyroxenes. Ti and Al contents of pyroxenes are very low and Ti, Cr, and Mn follow well-established magmatic differentiation trends. The high Cr content may reflect low  $f\text{O}_2$  conditions and/or early crystallization of olivine and plagioclase.

The  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios in lunar anorthosites are the lowest reported for any lunar rock. It is likely that anorthosites formed as cumulates during the major differentiation episode which occurred prior to  $\sim 4.3$  AE. Recrystallization features are common and  $^{39}\text{Ar}/^{40}\text{Ar}$  ages cluster around 4.0 AE. This may be the result of the intense bombardment prior to 4.0 AE which caused repeated cycles of in-situ fracturing and granulation followed by recrystallization. The low siderophile element content and the inferred slow cooling indicate a plutonic source region ( $>10$  km) not frequently plumbed by impact events. The Fe-rich silicates indicate crystallization from a melt at an advanced stage of fractionation. However, the low REE abundances are not consistent with late-stage crystallization. Plagioclase apparently crystallized relatively early and was concentrated by flotation and/or convection currents while the mafic minerals crystallized from a fractionated trapped liquid. The chemical, isotopic, and mineralogical data place stringent constraints on the nature of genetically related rocks and the relationship of anorthosites to other members of the ANT suite does not appear to be one of *simple* fractionation. The data presented in this paper are consistent with the Taylor-Jakes model of lunar evolution.

### 1. Introduction

Since the discovery of anorthositic clasts in Apollo 11 soils [1,2], it has been recognized that rocks of this type are an important component of the lunar highlands. Subsequent missions have served to emphasize this fact although it now appears that true anorthosites constitute a relatively small proportion of highland material exposed at the lunar surface. The

material sampled in highland regions, however, may not be representative of the lunar crust as a whole and there are indications that anorthosites may be more abundant in deeper crustal layers [3].

Anorthosites, i.e. rocks with more than 90 wt.% plagioclase, occur as lithic fragments in soil and rake samples, as clasts in polymict breccias, and as hand specimen sized samples, e.g. 15415, 60015, 60025, 61016 and 67075. Plagioclase crystal frag-

ments derived from anorthosites are undoubtedly present in soils and breccias. In addition, glasses with more than 90% normative plagioclase make up a significant fraction of lunar glasses [4]. However, anorthosites and anorthositic glasses are not uniformly distributed over the lunar surface. They are most abundant within highland terrane and along the Apennine front. Anorthosites constitute a small fraction of the soils at the Apollo 11 and Luna 16 sites but they are rare in the region of Oceanus Procellarum sampled by Apollo 12 and in the Fra Mauro Formation.

Lunar anorthosites show a number of distinctive features:  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios which are among the most primitive yet found in the solar system [5–7]; a low REE abundance pattern and a large positive Eu anomaly [8,9]; a very low siderophile element content [10]; a high FeO/MgO bulk rock ratio which reflects the Fe-rich compositions of olivines and pyroxenes; evidence for very slow cooling; textures indicative of extensive subsolidus annealing [11]. These features indicate that lunar anorthosites occupy a special position in lunar evolution. There can be little doubt that anorthosites, *sensu stricto*, are the products of very efficient crystal–liquid fractionation. Whether this fractionation took place during the crystallization of the hypothesized moon-wide magma system cannot yet be proved although it seems likely. However, the relationship of anorthosites to other highland rock types is not one of a *simple* fractionation process. In view of the importance of anorthosites, we have studied in detail the large, well-documented sample 67075.

## 2. Experimental

Petrographic and microprobe studies were made on polished thin sections 67075,18, 67075,47 and 67075,50 and on mineral fragments separated from bulk sample 67075,18. Microprobe and X-ray methods used have been described previously [12]. Initial studies showed that much of the detail of exsolution and inversion is obscured in precession photographs unless very small crystals ( $<200\ \mu\text{m}$ ) and a finely collimated beam (0.3 mm) are used. The small crystals required exposure times up to 400 hours.

## 3. General description

Sample 67075, an extremely friable cataclastic anorthosite (see fig. 7a in LSPET [13]), was collected at Station 11 on the rim of North Ray Crater near the contact of the Cayley Formation and the underlying Descartes Formation. The Cayley Formation in this region is estimated to be about 50 m thick and material was excavated to a depth of about 250 m [14]. A light-colored friable unit underlying the light-matrix breccias and overlying the darker breccia unit which forms the crater floor is the most likely *local* source for 67075. Krähenbühl et al. [10] believe that Apollo 16 cataclastic anorthosites represent deep ejecta from the Nectaris basin which was buried at a later date by shallow ejecta from younger ringed basins. 67075 has a well-defined  $^{40}\text{Ar}$ – $^{39}\text{Ar}$  plateau age of  $4.04 \pm 0.05\ \text{AE}$  [15] which may date the Nectaris basin-forming event. Plagioclase comprises over 90% of this sample, the remainder being olivine, low-Ca pyroxene, and high-Ca pyroxene, with trace amounts of Cr-spinel, ilmenite, Fe–Ni, and troilite [16–18]. The mafic minerals are not evenly distributed. In 67075,50, pyroxene is much more abundant than olivine, whereas 67075,18 shows the inverse relationship. The three published bulk analyses of 67075 [9,13,19] also indicate that the distribution of mafic minerals is not uniform, e.g. FeO varies from 1.07 wt.% to 3.41 wt.%. The mafic minerals tend to be concentrated in “veins”. The fragments in these “veins” have very similar compositions, indicating that a larger grain has been fragmented and extended into the vein-like feature. A coarse original grain size can occasionally be inferred from the relative orientation and shape of adjacent plagioclase fragments [20].

## 4. Mineralogy and textures

### 4.1. Plagioclase

Plagioclase occurs as (a) crystal fragments ranging in size from  $<1\ \mu\text{m}$  to  $\sim 2\ \text{mm}$ , (b) the major component in small rounded lithic clasts of microanorthosite, and (c) shocked and partially vitrified grains. The shock-free plagioclase grains are angular, complexly twinned (albite and pericline laws), and are not zoned.

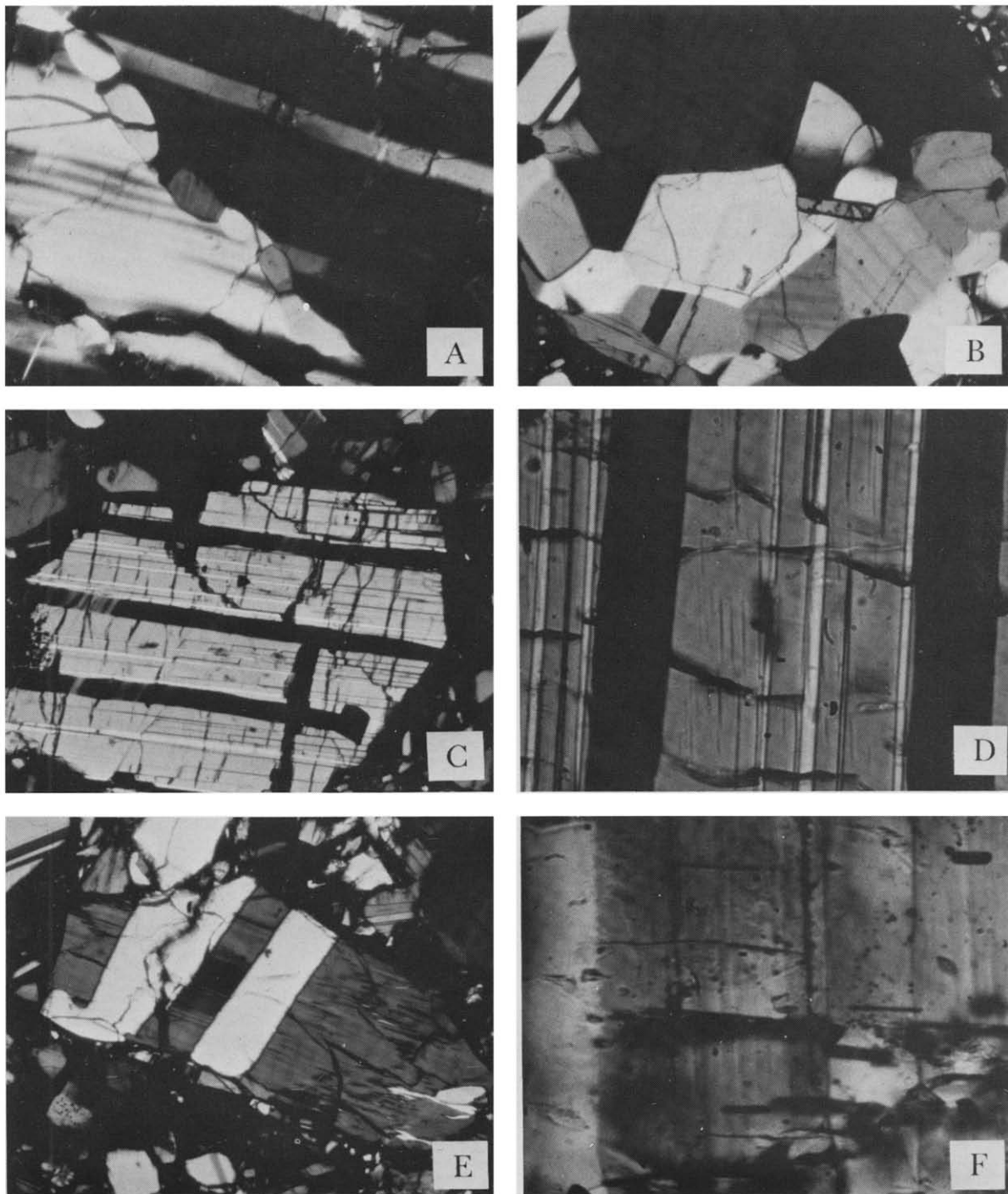


Fig. 1. Photomicrographs of 67075,50 illustrating textural and mineralogical features. (A–E: crossed polarized light; F: plane polarized light.)

A. Plagioclase crystal cut by a vein of small polygonal plagioclase grains which have healed a pre-existing fracture. Field of view is  $540 \times 430 \mu\text{m}$ .

TABLE 1

Microprobe analyses of plagioclases, olivines, pyroxenes, and spinel in 67075,18 and 67075,50\*.

|                                | 1      | 2      | 3      | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 11     |
|--------------------------------|--------|--------|--------|-------|-------|-------|-------|-------|-------|-------|--------|
| SiO <sub>2</sub>               | 43.89  | 44.13  | 33.36  | 34.94 | 50.36 | 51.10 | 51.13 | 51.28 | 51.29 | 51.18 | —      |
| TiO <sub>2</sub>               | 0.01   | 0.01   | 0.02   | 0.02  | 0.31  | 0.23  | 0.46  | 0.37  | 0.55  | 0.21  | 8.24   |
| Al <sub>2</sub> O <sub>3</sub> | 36.34  | 35.97  | 0.16   | 0.07  | 0.76  | 0.32  | 0.96  | 1.27  | 1.53  | 0.57  | 8.46   |
| Cr <sub>2</sub> O <sub>3</sub> | —      | —      | 0.02   | 0.02  | 0.31  | 0.15  | 0.31  | 0.30  | 0.33  | 0.30  | 40.77  |
| FeO                            | 0.06   | 0.05   | 47.63  | 38.81 | 25.03 | 31.14 | 14.44 | 18.05 | 13.30 | 30.08 | 39.01  |
| MnO                            | —      | —      | 0.58   | 0.46  | 0.46  | 0.55  | 0.27  | 0.38  | 0.29  | 0.47  | 0.44   |
| MgO                            | 0.02   | 0.02   | 18.19  | 25.39 | 13.41 | 14.65 | 10.73 | 13.08 | 11.90 | 16.25 | 1.46   |
| CaO                            | 19.71  | 19.55  | 0.13   | 0.09  | 7.81  | 1.65  | 21.24 | 15.02 | 20.71 | 0.83  | —      |
| Na <sub>2</sub> O              | 0.28   | 0.28   | —      | —     | —     | —     | —     | —     | —     | —     | —      |
| K <sub>2</sub> O               | 0.02   | 0.02   | —      | —     | —     | —     | —     | —     | —     | —     | —      |
|                                | 100.33 | 100.03 | 100.09 | 99.80 | 98.45 | 99.79 | 99.54 | 99.75 | 99.90 | 99.89 | 98.38  |
| <i>Cation proportions</i>      |        |        |        |       |       |       |       |       |       |       |        |
| O                              | 8.000  | 8.000  | 4.000  | 4.000 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 24.000 |
| Si                             | 2.024  | 2.039  | 0.991  | 0.992 | 0.990 | 0.999 | 0.983 | 0.982 | 0.975 | 0.992 | —      |
| Ti                             | 0.000  | 0.000  | 0.001  | 0.001 | 0.005 | 0.003 | 0.007 | 0.005 | 0.008 | 0.003 | 1.342  |
| Al                             | 1.975  | 1.959  | 0.006  | 0.002 | 0.018 | 0.007 | 0.022 | 0.029 | 0.034 | 0.013 | 2.158  |
| Cr                             | —      | —      | 0.001  | 0.001 | 0.005 | 0.002 | 0.005 | 0.005 | 0.005 | 0.005 | 6.979  |
| Fe                             | 0.002  | 0.002  | 1.183  | 0.922 | 0.412 | 0.509 | 0.232 | 0.289 | 0.212 | 0.488 | 7.062  |
| Mn                             | —      | —      | 0.015  | 0.011 | 0.008 | 0.009 | 0.004 | 0.006 | 0.005 | 0.008 | 0.081  |
| Mg                             | 0.001  | 0.001  | 0.805  | 1.075 | 0.393 | 0.427 | 0.308 | 0.373 | 0.337 | 0.470 | 0.470  |
| Ca                             | 0.974  | 0.968  | 0.004  | 0.003 | 0.165 | 0.035 | 0.437 | 0.308 | 0.422 | 0.017 | —      |
| Na                             | 0.025  | 0.026  | —      | —     | —     | —     | —     | —     | —     | —     | —      |
| K                              | 0.001  | 0.001  | —      | —     | —     | —     | —     | —     | —     | —     | —      |

\* Additional mineral analyses are listed in McCallum et al. [64].

1 = 67075,50 plagioclase (matrix); 2 = 67075,50 plagioclase (clast); 3 = 67075,50 olivine; 4 = 67075,18 olivine; 5 = 67075,18-14 pigeonite bulk analysis; 6 = 67075,18-14 pigeonite host; 7 = 67075,18-14 (001) augite lamella; 8 = 67075,18-4 augite bulk analysis; 9 = 67075,18-4 augite host; 10 = 67075,18-4 (001) pigeonite lamella; 11 = 67075,18-4 spinel lamella.

Healed fractures are a common feature of large plagioclases (Fig. 1A), and most grains show evidence of thermally-induced recrystallization, cf. 15415 [11]. Plagioclase is the dominant mineral in all lithic clasts: of the 30 lithic clasts observed in three sections, 24

are microanorthosites, 4 are noritic microanorthosites, and 2 are gabbroic microanorthosites (based on the classification scheme of Prinz et al. [21]). No troctolitic anorthosites were found. All microanorthosites show a well-developed polygonal grain

B. Lithic clast of microanorthosite containing one small crystal of orthopyroxene. Note the curved polygonal grain boundaries. Field of view is 540 × 430 μm.

C. Crystal of augite showing three broad (001) exsolution lamellae (20 μm) of partially inverted pigeonite (at extinction) and many narrow (001) lamellae (1–2 μm) of uninverted pigeonite. Very fine (100) pigeonite lamellae are visible in upper right corner. The bands in the broad pigeonite lamellae are parallel to (100) and may be related to the inversion or to second-generation exsolution of (100) augite. Field of view is 540 × 430 μm.

D. High-magnification photograph of central part of crystal shown in Fig. 1C. Note the three sets of “001” pigeonite (the wide lamellae are at extinction). Each set is slightly rotated with respect to the others. Field of view is 135 × 105 μm.

E. Broad (~50 μm) lamellae of augite on (001) of pigeonite host. Fine (100) lamellae are also visible. The pigeonite has almost totally inverted to orthopyroxene. Field of view 540 × 430 μm.

F. Minute rods of spinel exsolved from pyroxene. The spinel rods are oriented at ~90° to the trace of (001) and are present in both augite and pigeonite components. Field of view 135 × 105 μm.

boundary texture (Fig. 1B) typical of extensively annealed rocks.

There are no systematic compositional differences in the various types of plagioclase (Table 1 and Fig. 2A). The concentrations of minor elements (Mg, Fe, Ti, K) are significantly lower than those in plagioclases from mare basalts. However, the Fe/Fe + Mg ratio is high relative to that in other highland plagioclases (Fig. 2B) and comparable to that in plagioclases in 60025 [22]. Smith and Steele [23] have suggested that the high Fe/Fe + Mg ratio is not a primary feature but is the result of exsolution of pyroxene and silica from an initially non-stoichiometric plagioclase. It should be pointed out, however, that many large plagioclases in 67075 do not contain pyroxene and silica inclusions, nor is there any evidence to suggest that the plagioclases were initially non-stoichiometric. In addition, the large Fe-rich pyroxene and olivine grains which comprise between 5 and 10% of this sample are not the products of exsolution.

#### 4.2. Olivine

Olivine occurs as isolated, angular, and unzoned crystal clasts up to 0.5 mm in size. The most remarkable feature of olivines in 67075 is their high Fe content. A bimodal distribution of olivine compositions with maxima at  $Fa_{48}$  (range  $Fa_{45-50}$ ) and  $Fa_{59}$  (range  $Fa_{58-60}$ ) is observed (Fig. 5A, Table 1). Com-

parison with the voluminous microprobe data on highland olivines reveals that such Fe-rich olivines are extremely rare in highland rocks and soil.

If, as seems likely, these olivines were in equilibrium with a melt at some stage in their history, the coexisting melt phase would have a Fe/Fe + Mg (mole ratio) of 0.74 to 0.82 [24]. Such Fe-rich melts would represent the late-stage products of fractional crystallization. The high Fe/Fe + Mg ratios in plagioclases and pyroxenes reinforce this conclusion. The low minor element (Ca, Cr, Mn, and Ti) contents and compositional homogeneity clearly distinguish these highland olivines from Fe-rich olivines in mare basalts.

#### 4.3. Pyroxenes

High-Ca pyroxenes (augites) and low-Ca pyroxenes (pigeonites, partially inverted pigeonites, and orthopyroxenes) occur in approximately equal proportions as anhedral grains ranging in size from  $<1\ \mu\text{m}$  to 0.8 mm. Large matrix grains show exceptionally well-developed exsolution lamellae up to  $30\ \mu\text{m}$  wide [16, 25], while the smaller grains are relatively homogeneous or consist of portions of lamellar intergrowths (Fig. 1C–E). Pyroxenes in lithic clasts do *not* show the coarse exsolution textures, and some of these may be the result of exsolution from plagioclase. Large matrix fragments contain inclusions of chrome-rich spinel and ilmenite, most of which are equant and anhedral and not epitaxially related to the host pyroxenes. However, a small fraction of the spinel occurs as exsolved rods oriented in two directions (Fig. 1F), cf. Luna 20 pyroxenes [12]. It is not clear whether the non-oriented inclusions were incorporated during growth or exsolved and annealed. Rare anhedral grains of troilite, Fe–Ni and ilmenite and subspherical plagioclase ( $An_{97.2}$ ) are also found as inclusions in some pyroxene clasts.

##### 4.3.1. Augite.

The (*h0l*) precession photograph of a typical lamellar augite (crystal No. 4) is shown in Fig. 3A, the cell dimensions of its components are listed in Table 2, and the compositions of the bulk crystal and daughter products are given in Table 1. Fourteen reciprocal lattices can be identified on the (*h0l*) photograph of crystal No. 4, and our preferred interpretation of the thermal history is outlined schematically in Fig. 3B.

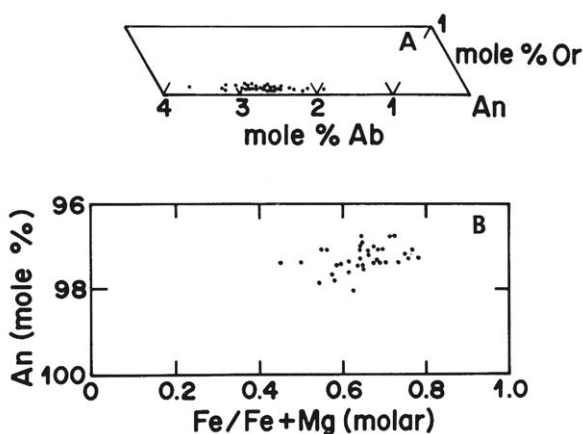


Fig. 2. Compositional variation in plagioclases from 67075,50 and 67075,18.

A. Or–Ab–An contents (normalized to 100%). B. An vs. Fe/Fe + Mg.

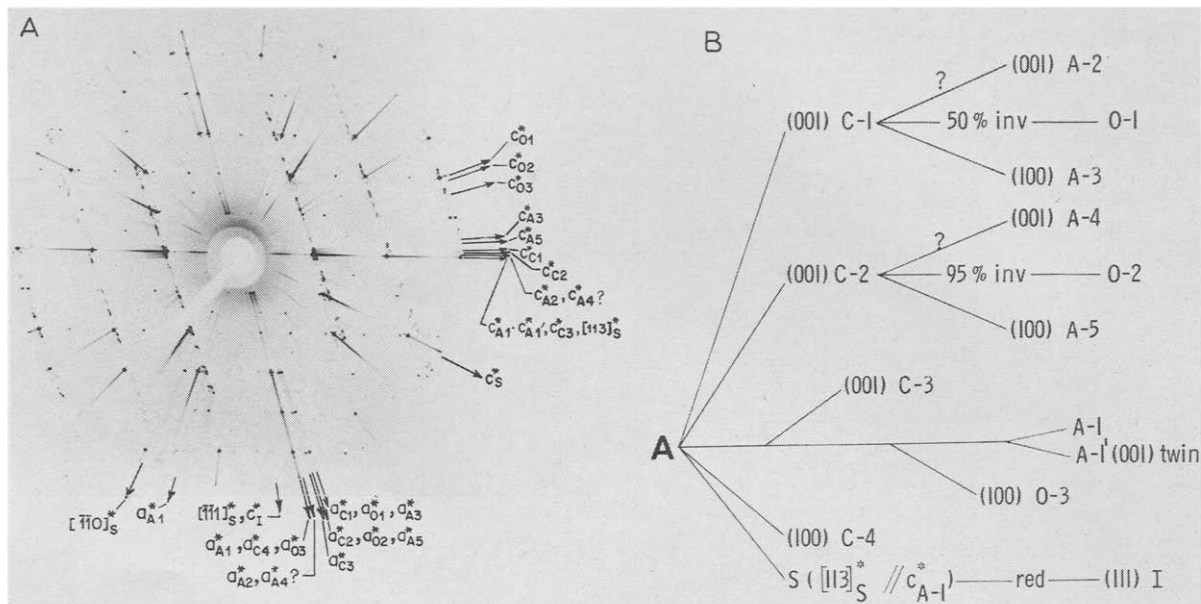


Fig. 3. X-ray precession photograph of the  $(h0l)$  net of augite No. 4. The reciprocal axes of the components are labeled  $a_{A1}^*$ ,  $c_{A1}^*$ , etc. Details are discussed in the text. A schematic thermal evolution diagram is shown on the right. Abbreviations: A = augite, C = pigeonite, O = orthopyroxene, S = spinel, I = ilmenite, Inv = inversion, Red = reduction, ? = inferred. Unless otherwise indicated, the arrows indicate exsolution.

TABLE 2

Cell dimensions of pyroxenes and spinel from 67075,18

| Crystal No. | Component              | $a(\text{\AA})$      | $b(\text{\AA})$ | $c(\text{\AA})$      | $\beta(^{\circ})$ | $\Delta c^*(^{\circ})^1$ |
|-------------|------------------------|----------------------|-----------------|----------------------|-------------------|--------------------------|
| 4           | A-1 host <sup>2</sup>  | 9.73(2)              | 8.92(2)         | 5.24(1)              | 105.98(8)         | 1.89                     |
|             | A-1' twin              | 9.72(2)              | 8.92(2)         | 5.25(1)              | 105.88(8)         |                          |
|             | C-1 (001) <sup>2</sup> | 9.66(2)              | 8.92(2)         | 5.21(1)              | 108.63(8)         |                          |
|             | O-1 (inv)              | 18.30(4)             | 8.92(2)         | 5.21(1)              | 90                |                          |
|             | A-3 (100)              | 9.75(2)              | 8.92(2)         | 5.21(1) <sup>3</sup> | 105.98(8)         | 1.00                     |
|             | C-2 (001) <sup>2</sup> | 9.64(2)              | 8.92(2)         | 5.22(1)              | 108.54(8)         |                          |
|             | O-2 (inv)              | 18.24(4)             | 8.92(2)         | 5.22(1)              | 90                |                          |
|             | A-5 (100)              | 9.75(2)              | 8.92(2)         | 5.22(1) <sup>3</sup> | 106.00(8)         |                          |
|             | C-3 (001)              | 9.73(2) <sup>3</sup> | 8.92(2)         | 5.21(1)              | 108.79(8)         | 0.00                     |
|             | C-4 (100)              | 9.64(2)              | 8.92(2)         | 5.24(1) <sup>3</sup> | 109.08(8)         |                          |
|             | O-3 (100)              | 18.22(4)             | 8.92(2)         | 5.24(1) <sup>3</sup> | 90                |                          |
|             | S (001)                | 8.31(2)              | 8.31(2)         | 8.31(2)              | 90                |                          |
| 14          | C-1 host <sup>2</sup>  | 9.67(2)              | 8.93(2)         | 5.21(2)              | 109.08(8)         | 1.33                     |
|             | A-1 (001) <sup>2</sup> | 9.73(2)              | 8.93(2)         | 5.25(2)              | 105.75(8)         |                          |
|             | C-3 (100)              | 9.64(2)              | 8.93(2)         | 5.24(2) <sup>3</sup> | 108.79(8)         |                          |
|             | A-2 (100)              | 9.74(2)              | 8.93(2)         | 5.21(2) <sup>3</sup> | 105.80(8)         |                          |
|             | O-1 (inv)              | 18.31(4)             | 8.93(2)         | 5.21(2)              | 90                |                          |
|             | O-1 (random)           | 18.3(1)              | 8.93(4)         | 5.21(4)              | 90                |                          |

<sup>1</sup>  $\Delta c^* = c_{A1}^* \wedge c_{C(001)}^*$  or  $c_{C(001)}^* \wedge c_{A(001)}^*$ .

<sup>2</sup> Chemical analyses of these components are listed in Table 1: A-1 host – analysis 9; C-1 (001) and C-2 (001) – analysis 10; C-1 host – analysis 6; A-1 (001) – analysis 7. Abbreviations explained in the text and caption to Fig. 3.

<sup>3</sup> Note the anomalous values.

The augite host (A-1) consists of a single crystal twinned on (001) with the twin units in the ratio of  $\sim 8:1$ , while the “001” pigeonite lamellae exist in three distinct lattice orientations (C-1, C-2, C-3). C-1 and C-2 do not share (001) with the augite host but make angles  $\Delta c^*$  ( $=c_{\text{aug}}^* \wedge c_{\text{pig}}^*$ ) of  $1.89^\circ$  and  $1.00^\circ$  respectively. Similar lattice rotations have been observed in terrestrial *plutonic* pyroxenes [26,17] and may indicate lattice relaxation to relieve strains induced by differential thermal contraction. Within the uncertainty of film measurement, the  $\beta$  angles of C-1 and C-2 are the same. C-3 has retained a near-perfect lattice orientation with respect to the augite host, i.e.  $\Delta c^* = 0^\circ$ . However, C-3 has an anomalously large  $a$  cell dimension (Table 2) which is identical to that of the host augite indicating lattice coherence across the interface. In our preferred interpretation, C-1 and C-2 are correlated with the broad (20  $\mu\text{m}$ ) “001” exsolution lamellae. An additional set of very fine (001) lamellae is visible microscopically (Fig. 1D) but reflections from these lamellae were not detected.

The three sets of “001” exsolution lamellae show different degrees of inversion to orthopyroxene: C-1 has partially inverted ( $\sim 50\%$ ) to orthopyroxene (O-1) sharing the (100) and (010) planes as indicated by the strong diffuse streaks parallel to  $a^*$  connecting equivalent reflections (Fig. 3A); C-2 has almost totally inverted to orthopyroxene (O-2) – very faint  $a^*$  diffuse streaks extend towards “relict” pigeonite reflections; C-3 shows no detectable inversion to orthopyroxene. C-1 and C-2 have exsolved second-generation augite on both (001) and (100) planes while C-3 shows no second-generation exsolution of augite.

The relationship between C-1 and C-2 is somewhat problematical. The faint diffuse streaks connecting equivalent reflections imply a genetic relationship. Reflection intensities indicate that C-1 and C-2 represent the volumetrically dominant “001” lamellae which make up 27% of crystal No. 4. However, C-2 shows significant exsolution of second-generation (100) augite coupled with almost total inversion whereas C-1 shows almost no (100) augite exsolution and only partial inversion suggesting that the extent of inversion is inversely proportional to the Ca content. In addition, much of the second-generation exsolution occurred prior to inversion. Ross et al. [28] have described a similar pattern of “001”

exsolution in an augite from the Bushveld complex.

Augite (A-1) has exsolved narrow lamellae of both pigeonite (C-4) and orthopyroxene (O-3) on (100). The absence of diffuse streaks parallel to  $a^*$  suggests that O-3 was exsolved after the crystal had cooled below the ortho  $\rightarrow$  clino inversion temperature and that it is *not* a product of inversion of C-4. Both C-4 and O-3 share  $a^*$  with the host augite and have  $c$  cell dimensions which are anomalously large and identical to that of the host augite.

The crystallographic relationship of the exsolved spinel is:  $[113]^*_S/c^*_{(A-1)}$  and  $[110]^*_S/b^*_{(A-1)}$ . Very faint reflections from ilmenite (I) can also be indexed. The (0001) plane of ilmenite is parallel to the (111) plane of spinel (S) suggesting that the two phases share close-packed oxygen layers. The ilmenite may have formed by subsolidus reduction of spinel or it may represent a separate exsolved phase.

The minor augite twin component (A-1') shows no exsolution features indicating that the twinning occurred after exsolution was complete. The orientation of the twin plane (001) is consistent with shock-induced twinning [29]. Similar features have been observed in augite from anorthosite 15415 [11].

#### 4.3.2. Low-Ca pyroxene

Most of the larger fragments of low-Ca pyroxene are inverted or partially inverted pigeonites with well-developed exsolution features (Fig. 1E). Many of the small matrix grains are orthopyroxene which, however, may show faint diffuse streaks parallel to  $a^*$  suggesting a monoclinic precursor. All low-Ca pyroxenes have  $\text{Fe}/(\text{Fe} + \text{Mg}) > 0.35$  which is consistent with initial (magmatic) crystallization as pigeonite.

Fig. 4A shows the ( $h0l$ ) precession photograph of a typical “pigeonite” (crystal No. 14). The cell dimensions of the component phases are listed in Table 2 and the compositions of those components resolvable by the microprobe are given in Table 1. A thermal evolution scheme is outlined in Fig. 4B.

After the exsolution of coarse augite plates (up to 30  $\mu\text{m}$ ) on (001) and narrow ( $\sim 1 \mu\text{m}$ ) lamellae of untwinned augite on (100) had depleted the host pigeonite in Ca, lattice adjustment occurred at the (001) lamellar interface ( $\Delta c^* = 1.33^\circ$ ) while the (100) lamellar interface remained coherent ( $\Delta a^* = 0^\circ$ ). The cell dimensions of (001) augite are normal while the  $c$  cell dimension of (100) augite is anomalously small

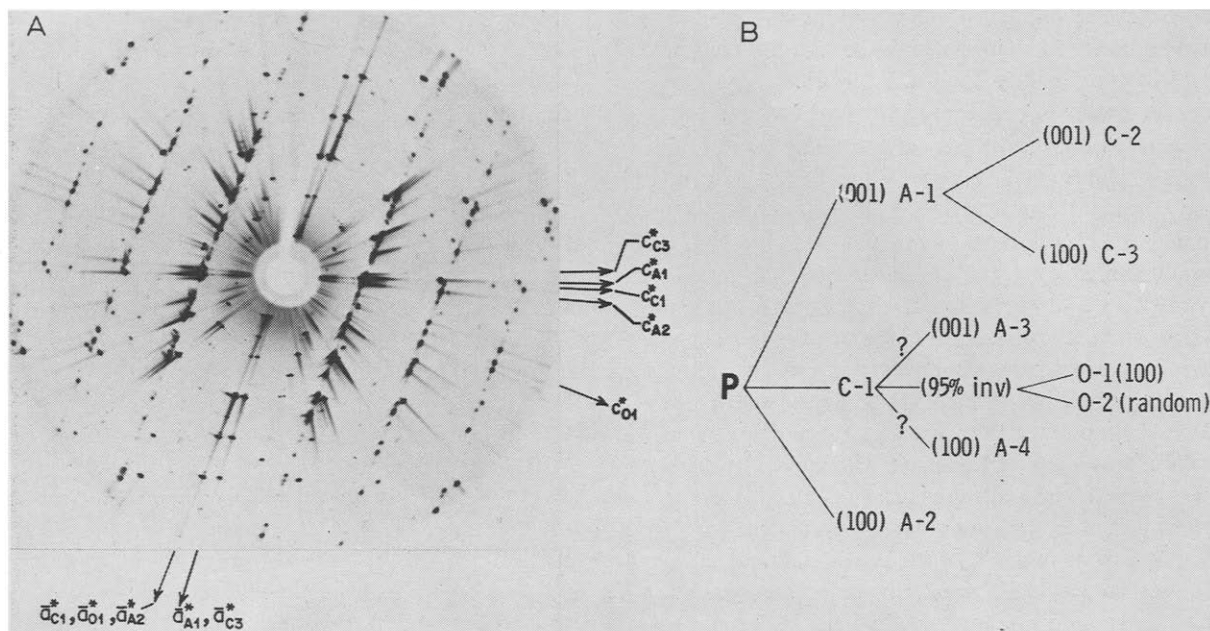


Fig. 4. X-ray precession photograph of the  $(h0l)$  net of pigeonite No. 14. Abbreviations are as in Fig. 3. Details are discussed in the text.

and identical to that of the host pigeonite (Table 2). Minor amounts of second-generation exsolution are observed (Fig. 4).

The pigeonite has largely inverted to orthopyroxene (~95%) with the two phases sharing the (100) and (010) planes. Diffuse streaks parallel to  $a^*$  are again indicative of intermediate states in the clino  $\rightarrow$  ortho transition. In addition, crystal No. 14 contains orthopyroxene with an irrational orientation. The reflections located principally between the  $(h02)$  and  $(h0\bar{2})$  lattice rows are from this second orthopyroxene which has an approximate orientation with respect to the (001) augite lamellae of  $[302]^*_{\text{Opx}} // c^*_{\text{A}}(001)$  and  $b^*_{\text{Opx}} // a^*_{\text{A}}(001)$ .

#### 4.3.3. Clino $\rightarrow$ ortho transition and lattice relaxation

Lattice relaxation to relieve strains is indicated by non-zero values of  $\Delta c^*$  and appears to be restricted to broad (001) exsolution lamellae formed under conditions of very slow cooling [27,30]. Narrow (001) and (100) lamellae commonly show anomalous  $a$  and  $c$  cell dimensions, respectively, indicating coherence

across the interface. Pigeonite lamellae which show anomalous cell dimensions remain uninverted.

The transition from clino to ortho occurs in both the broad "001" lamellae in augite and the host phase in pigeonite. In the former case, the orthopyroxene is oriented with respect to the monoclinic parent while in the latter case, the orthopyroxene may be oriented, non-oriented, or both (as in crystal No. 14). Randomly oriented orthopyroxene formed by inversion of pigeonite is commonly observed in slowly cooled pigeonites [30–32]. On the basis of experimental studies on the ortho  $\rightarrow$  clino transition, Ross et al. [32] suggest that irrational orientation may result from an intergranular recrystallization process in the presence of a melt phase. Since it is difficult to explain both oriented and non-oriented orthopyroxene in the same grain by this hypothesis, the possibility of oriented inversion with subsequent growth of orthopyroxene across grain boundaries cannot be ruled out.

The extent of the clino  $\rightarrow$  ortho transition in low-Ca pyroxenes is a potentially useful indicator of cooling rate. However, the transition temperature is a function of the Ca and Fe contents of the pyroxene

[32] and this will have an effect on the kinetics of the transition. In lamellar pyroxenes, the width and orientation of the exsolution lamellae affect the kinetics and it appears that lattice relaxation is required for the nucleation of orthopyroxene. As the examples discussed above make clear, it is possible to have a complete spectrum from 0% to 100% transition in a single grain. Until experimental data are obtained, deductions of cooling rate will remain qualitative.

Experimental studies in the "pyroxene system" [33,34] indicate a significant compositional gap between coexisting pigeonite and orthopyroxene on the stable three tie line or three phase tie triangle. These results imply that the inversion of pigeonite should involve a reaction such as pigeonite ( $\sim\text{Wo}_{10}$ )  $\rightarrow$  orthopyroxene ( $\sim\text{Wo}_{3-5}$ ) + augite ( $\sim\text{Wo}_{40-50}$ ). In every case which we have examined, pigeonite lamellae exsolved from augite have  $\text{Wo} < 5$  and it is not possible on the basis of composition to distinguish between pigeonite and inverted pigeonite lamellae. This suggests that exsolution proceeds to relatively low temperatures along the metastable extension of the pigeonite-augite solvus. In this case, the temperature at which inversion takes place cannot be specified. However, it appears that the metastable pigeonite solvus limb lies very close to the stable orthopyroxene limb and the two may even intersect as suggested by Ishii and Takeda [35].

#### 4.3.4. Chemistry

Chemical data on pyroxenes from 67075 are summarized in Figs. 5–8, and representative analyses are listed in Table 1.

Due to the coarse texture of the lamellar intergrowths, only a few grains were appropriate for bulk analysis. The five augite bulk analyses shown in Fig. 5A are of grains with multiple exsolution lamellae and are believed to be representative while the two pigeonite analyses were obtained on smaller grains and these may not be truly representative. Augites are less calcic and pigeonites more calcic than compositionally comparable pyroxenes from terrestrial plutonic intrusions. While there is a significant intergrain variation in  $\text{Ca} : \text{Fe} : \text{Mg}$  ratios, the lamellar components in individual grains are remarkably homogeneous (Fig. 6). Since the position of the solvus surface in  $T-X$  space is relatively insensitive to minor element variations [36], the compositions of plutonic igneous pyroxenes, and in particular their  $\text{Ca}$  contents, are primarily a function of the solidus temperature of the magma from which they are crystallizing. Such data as are available on the location of the solvus surface in the "pyroxene" system [33–37] indicate that the compositions of the bulk pyroxenes in 67075 are consistent with equilibrium plutonic crystallization at a higher temperature than comparable pyroxenes from terrestrial plutonic complexes. The alternate explana-

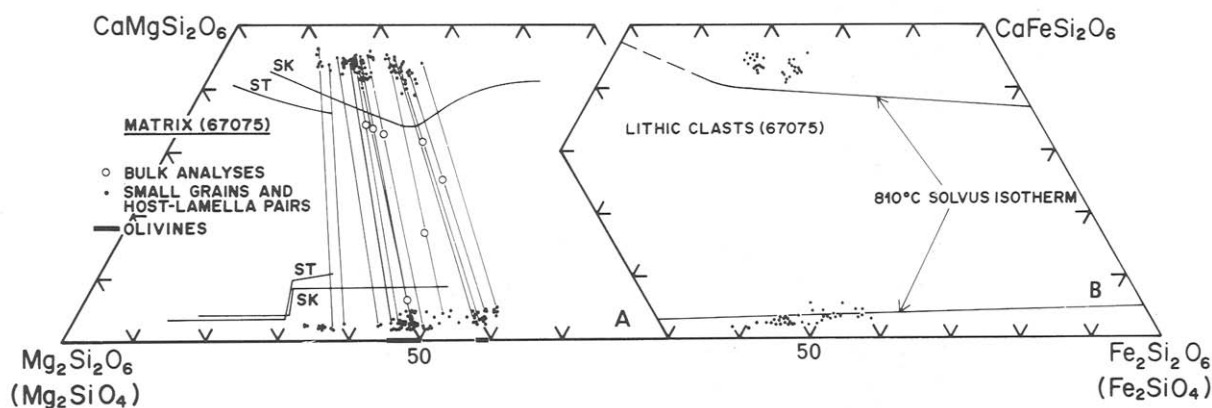


Fig. 5. Compositions of pyroxenes and olivines in 67075,50 and 67075,18 normalized to 100% ( $\text{Wo} + \text{En} + \text{Fs}$ ) and 100% ( $\text{Fo} + \text{Fa}$ ). A. Matrix pyroxenes. SK = trends of Skaergaard pyroxenes [42], ST = trends of Stillwater pyroxenes [41]. Tie lines connect adjacent host-lamella pairs. Note the tendency towards a bimodal distribution in olivine, low-Ca and high-Ca pyroxene compositions. B. Pyroxenes from lithic clasts. The 810°C isotherm is from Lindsley et al. [37].

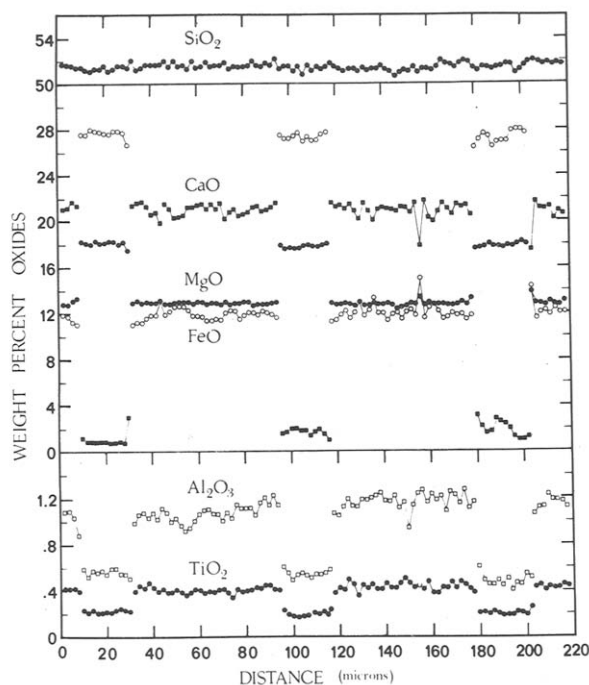


Fig. 6. Quantitative step scan profile (2- $\mu$ m steps) across the pyroxene shown in Fig. 1C.  $\text{Cr}_2\text{O}_3$  and  $\text{MnO}$  data are not plotted. Note the three different scales on the ordinate and also the homogeneity of the lamellar components.

tion that the pyroxenes crystallized metastably in a near-surface environment cannot be completely ruled out although it appears unlikely.

In contrast to the macrocrystals, the host and lamellae of exsolved pyroxenes together with small matrix grains and grains in lithic clasts show a wide compositional gap between the high-Ca and the low-Ca varieties and a greater range in  $\text{Fe}/\text{Fe} + \text{Mg}$  (Figs. 5A,B and 6). However, individual grains are homogeneous. The scatter of points in Fig. 5 tends to obscure the bimodal distribution of compositions; the majority of the analyses cluster around  $\sim\text{Wo}_{44}\text{En}_{37}\text{Fs}_{19}$  and  $\sim\text{Wo}_3\text{En}_{50}\text{Fs}_{47}$  and a minority around  $\sim\text{Wo}_{43}\text{En}_{31}\text{Fs}_{26}$  and  $\sim\text{Wo}_4\text{En}_{39}\text{Fs}_{57}$ . Mg-rich pyroxenes constitute a small ( $\sim 1\%$ ) but nonetheless significant component. The mean compositions of the major group are similar to those of pyroxenes in anorthosites 60025 [22,38], 61016 [39], 15415 [11,40], Luna 20 [21], and the 1–2-mm fraction of Apollo 15 and 16 soils (G.J. Taylor, personal communication). The high iron content of pyroxenes

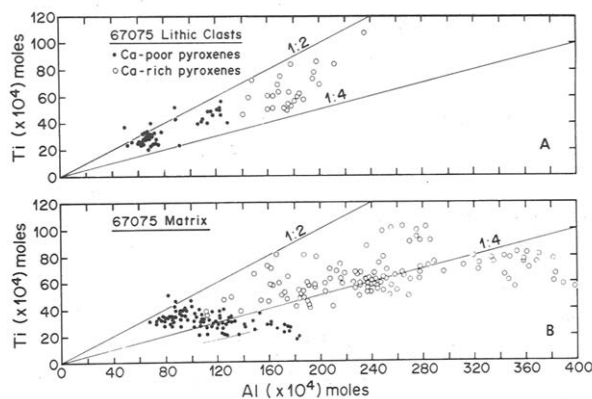


Fig. 7. Ti–Al relationships of pyroxenes in 67075. The data are normalized to three oxygen atoms and are not corrected for the substitution of  $\text{CaCrAlSiO}_6$ . A. Lithic clasts. B. Matrix. Note the very low Ti and Al contents in both, and the greater range of matrix pyroxenes.

in cataclastic anorthosites, particularly 67075, contrasts with the Mg-rich pyroxenes in most highland samples.

Comparison with the 810°C isotherm on the orthopyroxene–augite solvus surface [37] clearly shows that intracrystalline diffusion in pyroxenes from 67075 was effective on a micron scale to temperatures less than 800°C (Fig. 5B). In addition, the  $K_D$ 's ( $K_D = X_{\text{Fe}}^{\text{opx}} X_{\text{Mg}}^{\text{cpx}} / X_{\text{Mg}}^{\text{opx}} X_{\text{Fe}}^{\text{cpx}}$ ) of numerous adjacent host-lamellae pairs varies from 1.50 to 1.90 with a mean of 1.68. This indicates “equilibration” to temperatures as low as 700°C to 800°C [41].

The Ti–Al relationships are shown in Fig. 7A (lithic clasts and 7B (matrix). Although there is some overlap in the data, matrix pyroxenes show a much greater variation in Ti/Al ratios and significantly different trends; all pyroxenes from lithic clasts have  $\text{Ti}/\text{Al} > 1/4$ . The Ti and Al contents of all pyroxenes are very low, the only comparable pyroxenes being those from other cataclastic anorthosites, e.g. 60025 [38]. Ti in pyroxenes increases as  $\text{Fe}/\text{Fe} + \text{Mg}$  increases during progressive fractional crystallization of basaltic magma [41,42], as long as Ti-rich oxide phases are not crystallizing simultaneously. Matrix pyroxenes from 67075 follow such a trend (Fig. 8A). In their discussion of lunar pyroxenes, Taylor et al. [43] indicate that Ti decreases with progressive fractionation. However, they make no distinction among orthopyroxene, pigeonite and augite nor do they differen-

tiate pyroxenes from widely disparate rock types. This tends to obscure the true relationships. Pyroxenes from 67075 have  $\text{Ti}/\text{Al} < 1/2$  indicating the presence of octahedrally coordinated aluminum. This conclusion remains valid when the data are adjusted for the substitution of Cr. In addition, a plot of Ti versus  $2(1-\text{Si})-(\text{Cr} + \text{Al}-\text{Na})$  reveals the presence of small amounts of  $\text{Ti}^{3+}$ .

The Cr content of pyroxenes from 67075 is high relative to compositionally comparable pyroxenes from terrestrial layered complexes (Fig. 8B) but not anomalously high when compared to pyroxenes from other ANT rocks [43–45]. The exsolution of chrome spinel reflects this high Cr content. During fractional crystallization of basic melts, particularly those precipitating pyroxenes, the Cr content of the residual melt decreases dramatically (Fig. 8B) as a result of the strong octahedral site preference of Cr. This has been repeatedly verified for lunar rocks although the depletion in Cr with progressive fractional crystallization does not appear to be as extreme as observed in terrestrial examples. The scatter of points in Fig. 8B is due in part to the minute spinel lamellae and precludes any conclusions as to variation of Cr with  $\text{Fe}/\text{Fe} + \text{Mg}$  in this case.

There are several possible explanations of the high Cr content of ANT pyroxenes relative to terrestrial equivalents. In the first place, the magmatic system from which the ANT suite formed by differentiation was enriched in refractory elements, including Cr [3]. This in itself is not sufficient to explain the high Cr contents of late differentiates. It is also necessary to postulate the early crystallization of Cr-poor phases such as olivine and plagioclase. Alternatively, the low  $f\text{O}_2$  under which lunar rocks formed would be reflected in increased  $\text{Cr}^{2+}/\text{Cr}^{3+}$  ratios. The presence of  $\text{Cr}^{2+}$  in lunar silicates has been suggested [46–48], but spectral evidence for  $\text{Cr}^{2+}$  is lacking [49]. The geochemical behavior of  $\text{Cr}^{2+}$  is not well understood but according to Burns [50],  $\text{Cr}^{2+}$  may be concentrated in the liquid relative to the crystalline phases. If so, the relatively high Cr content of late differentiates may simply be a reflection of the low oxidation state of the magma. It is clear, however, that additional experimental data are required.

Mn is similar to that in other ANT pyroxenes and follows the well-established trend of enrichment with increasing  $\text{Fe}/\text{Fe} + \text{Mg}$  (Fig. 8C).

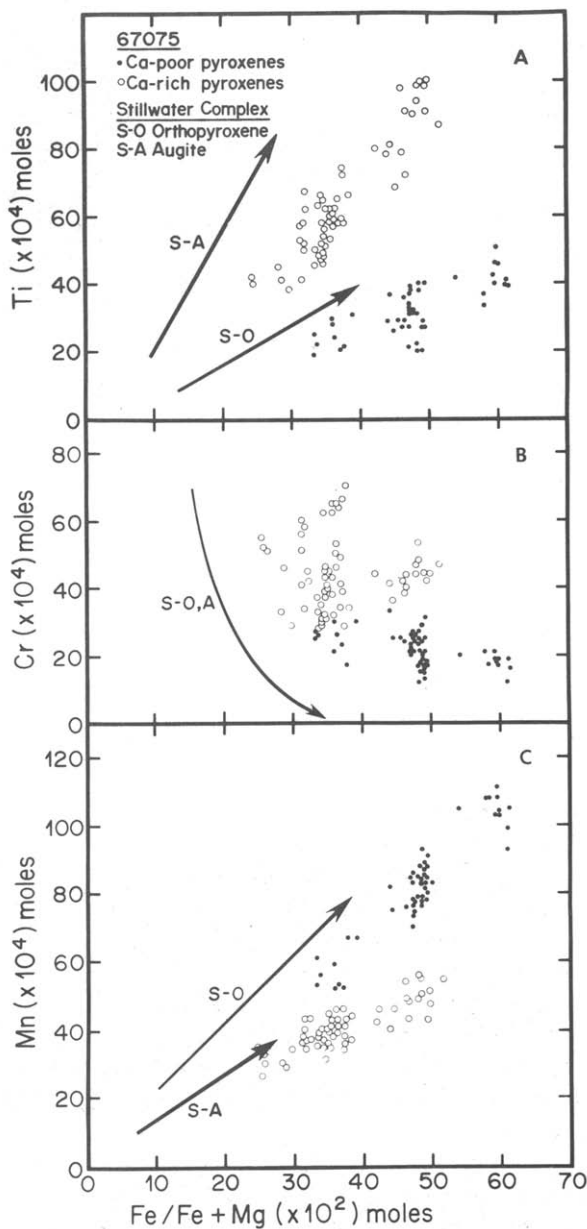


Fig. 8. Ti, Cr and Mn (normalized to three oxygen atoms) in pyroxenes from 67075 plotted against  $\text{Fe}/\text{Fe} + \text{Mg}$ . A. Ti. The trends for high-Ca and low-Ca pyroxenes from the Stillwater complex [41] are shown for comparison. At comparable degrees of Fe-enrichment, Stillwater pyroxenes are richer in Ti. In both cases Ti increases with progressive fractionation. B. Cr. The trends in pyroxenes from 67075 are not well-defined. The Cr content of Stillwater pyroxenes decreases rapidly and at comparable degrees of Fe-enrichment is much lower. C. Mn. Mn increases with progressive fractionation in both cases.

#### 4.4 Opaque minerals

The opaque minerals in 67075 have been discussed in detail by El Goresy et al. [18]. In addition to the primary and reduced spinel types found by these authors, the minute exsolved platelets of spinel constitute a third type. The interpretation that the spinel associated with ilmenite, Fe metal, and troilite has undergone intense subsolidus reduction is open to question. The analyses reported by El Goresy et al. [18] indicate no significant difference in composition between the primary and reduced spinels. If reduction is by the reaction  $(\text{Fe}_2\text{TiO}_4)_{\text{ss}} \rightarrow \text{Fe} + \text{FeTiO}_3 + 1/2 \text{O}_2$ , the reduced spinel should be depleted in the ulvospinel component. The occurrence of troilite with the reduced spinel is also difficult to explain by this mechanism.

### 5. Discussion

#### 5.1. Evidence for mixing

The non-uniform distribution of mafic minerals and lithic clasts in sample 67075 indicates that this rock is a polymict breccia. The three distinct components recognized by Smith and Steele [23] are consistent with this interpretation. In addition, the range in Fe/Fe + Mg ratios in the mafic minerals is larger than one would predict for a hand specimen-sized sample of plutonic igneous rock, cf. terrestrial layered intrusions. A final piece of evidence in favor of mixing comes from phase equilibrium considerations. Experimental studies in the systems Fo–Fa–An–SiO<sub>2</sub> [51] and MgO–FeO–Fe<sub>2</sub>O<sub>3</sub>–SiO<sub>2</sub>–An [52] indicate that, at low pressures in anorthite-saturated melts, there is a reaction relationship between olivine and low-Ca pyroxene. It follows that with either equilibrium ranges of olivine and pigeonite should not overlap to any significant extent. This is clearly not the case with sample 67075.

However, it appears from both crystallographic and chemical evidence that the olivine and pyroxene grains in the matrix of 67075 are genetically related and quite distinct from olivines and pyroxenes in other Apollo 16 breccias and soils. The origin of the small pyroxenes in the recrystallized lithic clasts is not clear; they have the same Fe/Fe + Mg range as the

matrix grains (Fig. 5), but the minor element contents are distinct (Fig. 7). It has been suggested that pyroxenes in lithic clasts have exsolved from plagioclase [22, 23] but this remains unproven. It is at least equally probably that these pyroxenes were formed by the recrystallization of fragmented primary grains.

Peckett and Brown [25] believe that the mineralogical features observed in 67075 are consistent with an origin by fragmentation and mixing of the components of a layered plutonic complex. We concur with this view and believe that a good analogy can be made with the rocks in the anorthosite zone of the Stillwater complex. The anorthosite zone in this layered complex contains a spectrum of rock types from pure anorthosites with no mafic minerals to gabbroic anorthosites with up to ~30% postcumulus pyroxene (augite + pigeonite or orthopyroxene). From the base to the top of this zone, there is a considerable variation in the Fe/Fe + Mg ratio in the pyroxenes [41]. Thin troctolite layers occur at several horizons within the anorthosite zone. While the olivines in any one troctolite unit are homogeneous, there is a distinct compositional difference in olivines from different units. If such a body were fragmented, crushed, mixed, and partially recrystallized, the resulting breccia would show many of the features observed in rock 67075. Reasoning by analogy is open to criticism but current knowledge of structural relationships among lunar highland rocks is such that more conventional avenues of reasoning are not appropriate.

Data obtained on terrestrial layered anorthosites also have a bearing on the apparent inverse correlation of An content in plagioclase and Fe/Fe + Mg ratio in coexisting mafic minerals which has been reported for rocks of the ANT suite [17]. Detailed analysis of coexisting minerals in the anorthosite zone of the Stillwater complex reveals numerous reversals of the predicted trend. The formation of cumulates is a complex process and the composition of coexisting phases is a function not only of the magma composition but also of the rate of cooling, differential settling or flotation rates, extent of reaction of cumulus crystals with intercumulus liquid, and the extent of subsolidus reequilibration. It seems premature at this point to place undue emphasis on this apparent anti-correlation trend. Dowty et al. [53] also conclude from their study of Apollo 16 rake samples that there is little justification for postulating such a trend.

TABLE 3

 $^{87}\text{Sr}/^{86}\text{Sr}$  ratios and  $^{40}\text{Ar}$ - $^{39}\text{Ar}$  ages of lunar anorthosites

| Sample No. | $^{87}\text{Sr}/^{86}\text{Sr}^*$ | Ref. | $^{40}\text{Ar}$ - $^{39}\text{Ar}$ age (AE) | Ref. |
|------------|-----------------------------------|------|----------------------------------------------|------|
| 15415      | $0.69896 \pm 4$                   | [5]  | $3.95 \pm 0.07$                              | [54] |
| 60015      | $0.69884 \pm 4$                   | [7]  | $3.55 \pm 0.05$                              | [55] |
| 60025      | $0.69893 \pm 3$                   | [56] | $4.18 \pm 0.06$                              | [55] |
|            | $0.69902 \pm 6$                   | [7]  |                                              |      |
| 61016      | $0.69888 \pm 3$                   | [6]  | $3.65 \pm 0.04$                              | [54] |
|            | $0.69894 \pm 3$                   | [57] |                                              |      |
| 67075      | —                                 |      | $4.04 \pm 0.05$                              | [15] |

\* The  $^{87}\text{Sr}/^{86}\text{Sr}$  values are corrected for interlaboratory bias relative to NBS standard SRM-987 and recalculated to the values at 4.6 AE.

### 5.2. Isotopic composition and ages of lunar anorthosites

The  $^{87}\text{Sr}/^{86}\text{Sr}$  values and  $^{39}\text{Ar}$ - $^{40}\text{Ar}$  ages of large anorthosite samples are summarized in Table 3. The spread in Ar ages is compatible with the petrographic features observed in cataclastic anorthosite, e.g. the simple Ar release pattern and well-defined plateau obtained for sample 67075 [15] correlates well with the observed recrystallization and the virtual absence of shock features. The plateau age of  $4.04 \pm 0.05$  AE most likely dates the recrystallization event. On the other hand, sample 60025 shows a complex release pattern and the plateau age of  $4.18 \pm 0.06$  AE is based only on the fraction released at high temperature; 60025 shows both recrystallization textures and the effects of moderate to heavy shock [22]. The occurrence of anorthosites with ages ranging up to 4.18 AE (4.26 AE if data obtained on anorthositic fragments in Apollo 16 soil [58] are included) has important implications for the evolution of the lunar crust. On the assumption that these dates represent recrystallization ages, it can be inferred that plagioclase-rich cumulates existed prior to 4.26 AE. Rb and Sr may be redistributed on a local scale during metamorphism but this would have a negligible effect on  $^{87}\text{Sr}/^{86}\text{Sr}$  ratios in anorthosites which are essentially free of phases with high Rb/Sr.

Rb—Sr model ages (relative to BABI) of soils and whole rocks indicate that the moon underwent a large-scale differentiation between 4.6 and 4.3 AE [59]. An important question is: Were the anorthosites formed initially during this event? In the absence of

(a) precise knowledge of the  $^{87}\text{Sr}/^{86}\text{Sr}$  value at the time of condensation of lunar material from the solar nebula, or (b) anorthosites with crystallization ages close to 4.5 AE, an unambiguous answer is not possible. On the assumption that the oldest Ar age ( $\sim 4.2$  AE) represents the time of primary formation of anorthosites, a value of  $(^{87}\text{Sr}/^{86}\text{Sr})_I$  at 4.6 AE of 0.69879 is obtained assuming  $(^{87}\text{Rb}/^{86}\text{Sr})_{\text{moon}} = 0.03$  [60]. While this is a reasonable value it is not compatible with the data obtained on dunite clast 72417 which has an age of  $4.6 \pm 0.09$  AE and  $(^{87}\text{Sr}/^{86}\text{Sr})_I = 0.69899 \pm 5$  [61]. According to Albee et al. [61], this dunite “...must represent a cumulate formed during the early lunar differentiation and associated differential gravitational settling.” This implies the existence at 4.6 AE of a large magmatic system with  $I = 0.69899 \pm 5$ . It appears more than coincidental that the cataclastic anorthosites give the same  $I$  values (C.I.T. data recalculated to 4.6 AE) within experimental uncertainty. On the basis of this admittedly circumstantial evidence, it is *tentatively* concluded that the cataclastic anorthosites represent primitive crustal material formed during the major differentiation episode which occurred shortly after accretion of the moon.

### 5.3. Depth of origin

The absence of meteoritic contamination in Apollo 16 anorthosites led Krähenbühl et al. [10] to conclude: “...the cataclastic anorthosites... must have originated in a deep stratum well below the ancient regolith, uncontaminated by projectile debris.” The

depth to which mixing and contamination were effective during the period of intense bombardment has been estimated at  $\sim 10$  km [3]. The large basin-forming events probably excavated material from depths of 40–60 km [3] but these events were sufficiently few in number that they did not contribute significantly to the fine-scale mixing. The relative rarity of anorthosites, in itself, suggests a source region not frequently plumbed by impact events. The evidence for an original coarse texture is also suggestive of formation at depth.

The formation of anorthositic cumulates requires conditions of very slow cooling. In this regard, it is worth noting that anorthositic layers of large horizontal and vertical dimensions are found only on the largest of the terrestrial layered complexes, e.g. Stillwater, Bushveld, and Duluth complexes. This is consistent with exsolution, inversion, and lattice relaxation features observed in pyroxenes from 67075 which are on a scale and of a type analogous to those in Stillwater pyroxenes [30]. It is of dubious value to calculate settling and cooling rates until more precise data are available on the dimensions of the magma system and on the composition, density, viscosity, and thermal properties of the magma from which the anorthosites were derived. In addition, it is necessary to evaluate the relative contributions of such heat sources as decay of K, Th, and U, electrical heating, and the conservation of impact energy. However, rough calculations clearly indicate that for a magma system of the postulated dimensions (100–500 km thick), cooling rates are sufficiently slow to permit effective accumulation of precipitated minerals even when  $\rho_{\text{magma}} - \rho_{\text{crystal}}$  is very small.

#### 5.4. Anorthosites as late-stage differentiates

The high Fe/Fe + Mg ratios in olivines and pyroxenes from lunar anorthosites [51,53] indicate crystallization from a melt with Fe/Fe + Mg between  $\sim 0.7$  and  $0.8$  [24]. The bulk moon has an Fe/Fe + Mg (mole ratio) of  $0.16$  [3]. Assuming an initial magma with this value, it can be shown that  $>95\%$  fractional crystallization is required to produce the Fe/Fe + Mg value calculated for the residual liquid in equilibrium with anorthosites.

The REE abundances can also be used to estimate the degree of fractionation. From analytical data on

67075 [9] and the experimentally determined distribution coefficients of Weill et al. [62], the Sm/Eu ratio of a liquid in equilibrium with plagioclase from 67075 is  $5.2$  and the concentration of Sm in the liquid is  $\sim 12 \times$  chondrites. With Sm abundances in the parent magma of  $5 \times$  chondrites [3], a  $12 \times$  Sm enrichment corresponds to  $65\%$  solidified for the fractional crystallization path 234E discussed by Weill et al. [62]. The degree of Sm enrichment is largely independent of the crystallization path in the system Fo–Fa–SiO<sub>2</sub>–An.

Sample 67075 shows higher REE abundances and a flatter trend than other large samples of anorthosite. These features may be explained by the presence of a significant amount of augite in 67075. The REE abundances in other anorthosites do not appear to be compatible with late-stage crystallization. Calculation of the REE of liquids in equilibrium with anorthosites give values of  $4 \times$  (15415, 60015, 61016) to  $8 \times$  (60025) chondrites. These values are not markedly different from those assumed for the whole moon [3]. Assuming an initial REE value of  $5 \times$  chondrites [3] for the magma, crystallization (and removal) of  $50\%$  by volume as olivine and orthopyroxene produces a residual magma with REE  $\sim 10 \times$  chondrites [62] ( $\sim 8 \times$  if augite crystallizes instead of orthopyroxene). However, crystallization of  $50\%$  mafics, which is about the maximum permissible to prevent excessive REE enrichments in the residual liquid, is not sufficient to generate the Fe/Fe + Mg ratio in the liquid required to produce the mafic minerals found in anorthosites. Adequate explanations of this paradox have not yet been proposed.

#### 5.5. Relationship to other highland rock types

Rocks from the lunar highlands have fallen victim to the acronym syndrome and it has become common practice to consider anorthosites together with plagioclase-rich norites and troctolites in an ANT suite, the implication being that members of the suite have been derived from a common parental magma by a process of crystal–liquid fractionation accompanied by accumulation of the crystalline phases [21,51]. This hypothesis is very attractive but it should be pointed out that it is based largely on a major element chemical data obtained on lithic clasts, soils, glasses, and polymict breccias. Remarkably few large monomict

rocks have been recognized within this suite and samples showing the textural features normally associated with cumulates are ever rarer. (Norite 78235 is the only large rock known to use which has retained a cumulate texture *and a grain size* consistent with plutonic crystallization). However, despite the mechanical and thermal degradation, a strong case can be made for considering anorthosites as plutonic igneous cumulates.

Walker et al. [51] and Prinz et al. [21] have discussed in detail the origin of the ANT suite by igneous fractionation. However, it was clearly recognized by Walker et al. [51] that a number of features of ANT rocks cannot be explained in a convincing manner by a simple fractionation scheme, and they cite, in particular, conflicting differentiation indices: anorthosites are quartz-normative and contain Fe-rich mafic minerals implying that they are late-stage differentiates while the calcic nature of the plagioclase (relative to other ANT samples) implies the opposite. Attempts to explain this anomaly by Na volatilization lack supporting evidence. As discussed earlier, it is not clear whether these apparently conflicting features are real. However, lunar anorthosites show additional features which tend to set them apart from other highland rocks.

Lunar anorthosites have large positive Eu anomalies expected for plagioclase-rich cumulates. However, as pointed out by Haskin et al. [9], other plagioclase-rich soils and breccias from interior highland terrane tend to show negative Eu anomalies. Under conditions of low  $fO_2$ , Eu (as  $Eu^{2+}$ ) is strongly fractionated relative to Sm into plagioclase. At  $1200^\circ C$ ,  $(Eu/Sm)_{\text{plag}} / (Eu/Sm)_{\text{melt}} \sim 30$  [62]. In general, plagioclase-rich cumulates will show a positive Eu anomaly. With continued removal of plagioclase, the residual liquid develops an increasingly large negative Eu anomaly. A point is reached during the later stages of fractional crystallization at which even plagioclase-rich cumulates will show a negative Eu anomaly. However, there is no compelling reason to believe that the members of the ANT suite which show a negative Eu anomaly are late-stage differentiates. In fact the  $Fe/Fe + Mg$  ratio of the mafic minerals implies the opposite.

Anorthosites are essentially free of meteoritic contaminants as revealed in their siderophile element content [10]. In contrast, other members of the ANT suite show variable, but generally high, siderophile

element contents. Anorthosites were apparently isolated from the "zone of mixing" during the intense early bombardment phase of lunar history.

The mafic minerals in anorthosites are chemically and crystallographically distinctive. Coarse exsolution textures are relatively common in pyroxenes from anorthosites but rare in other ANT rocks, indicating significantly different thermal histories. The low Ti content of pyroxenes in anorthosites is particularly significant since Ti tends to vary sympathetically with  $Fe/Fe + Mg$  during fractional crystallization. It is difficult to avoid the conclusion that pyroxenes in anorthosites crystallized under different conditions than those in the bulk of highland rocks.

It seems clear that the magmatic fractionation hypothesis as currently formulated for the origin of the ANT suite requires significant modification. While space limitations do not permit a detailed discussion in this paper, we shall briefly outline a model based in large part on the comprehensive Taylor-Jakes [3] model of lunar evolution. In the Taylor-Jakes model, at  $\sim 4.4$  AE the upper few kilometers of the lunar crust were composed of a relatively unfractionated assemblage of rocks derived by rapid cooling of the primitive moon-wide magma system. This assemblage is compared to the chilled margin of terrestrial intrusions. However, as recent calculations on the mechanism of consolidation of layered intrusions shows [63], the "chilled margin" is likely to remain relatively thin ( $< 1$  km) as heat lost by conduction through the crust is compensated by the upward transfer of heat by convection. A more satisfactory model for the consolidation of the upper part of the lunar crust involves a process similar to that which produced the Upper Border Group of the Skaergaard intrusion [42]. With increasing depth below the thin, chilled crust, the cooling rate decreased and fractionation became increasingly effective resulting in an increase with depth of both the plagioclase content and the  $Fe/Fe + Mg$  ratio in coexisting minerals. At this point the melt composition lay on the plagioclase-mafic mineral field boundary. Plagioclase was concentrated by flotation and/or transportation in convection currents which also act as effective stirrers. Because of the high density of the mafic minerals relative to that of the melt, the only mafics retained would be those crystallized from a liquid trapped in a mush of plagioclase crystals. With progressive crystallization, the density

of the melt increased slightly. This permits more efficient accumulation of plagioclase (crystal/pore liquid ratio increases) while the slower cooling permits more extensive adcumulus overgrowth.

During the early stages of crystallization, breccias in the crust caused by impacts would be filled with fractionated magma to a level controlled by isostatic equilibrium. In this way a range of fractionated rock types could form at the surface having compositions controlled by the plagioclase–mafic mineral cotectic. This effect would become increasingly small as the crust thickened and the impact rate declined.

An additional component of surface material would be feldspathic volcanic rocks (the Fra Mauro basalt suite) developed at a later time by internal partial melting of plagioclase-bearing cumulates. While volcanic rock types may not be abundant in the crust as a whole, they must have formed a significant fraction of the ancient regolith. Prior to the “cataclysm”, the terra regolith was composed of a well-mixed assemblage of “upper border group” rocks (troctolites, norites, anorthositic norites, noritic anorthosites), feldspathic volcanics (KREEP to high alumina basalts), and a small amount of meteoritic debris, with the zone of mixing extending to ~10 km.

The cataclysmic basin-forming events excavated material from deeper levels (40–60 km). Anorthosites are the most easily recognizable component of this plutonic material. Much of the deep material would fall back into the crater to be later covered by mare basalt flows while the remainder was ejected and partially mixed with the preexisting regolith. Bombardment decreased markedly after the basin-forming events and the plutonic material was not thoroughly mixed and remained essentially uncontaminated with meteoritic debris. The Apollo 16 cataclastic anorthosites were probably excavated by the Nectaris impact [10] and covered at a later stage by shallow-level ejecta derived from the distant Crisium, Imbrium, and Orientale basins.

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## References

- 1 J.A. Wood, J.S. Dickey, U.B. Marvin and B.N. Powell, Lunar anorthosites and a geophysical model of the moon, *Geochim. Cosmochim. Acta, Suppl.* 1, 1 (1970) 965–988.
- 2 J.V. Smith, A.T. Anderson, R.C. Newton, E.J. Olsen, P.J. Wyllie, A.V. Crewe, M.S. Isaacson and D. Johnson, Petrologic history of the moon inferred from petrography, mineralogy, and petrogenesis of Apollo 11 rocks, *Geochim. Cosmochim. Acta, Suppl.* 1, 1 (1970) 897–925.
- 3 S.R. Taylor and P. Jakes, The geochemical evolution of the moon, *Geochim. Cosmochim. Acta, Suppl.* 5, 2 (1974) 1287–1305.
- 4 W.I. Ridley, A.M. Reid, J.L. Warner, R.W. Brown, R. Gooley and C. Donaldson, Glass compositions in Apollo 16 soils 60501 and 61221, *Geochim. Cosmochim. Acta, Suppl.* 4, 1 (1973) 309–321.
- 5 G.J. Wasserburg and D.A. Papanastassiou, Age of an Apollo 15 mare basalt: lunar crust and mantle evolution, *Earth Planet. Sci. Lett.* 13 (1971) 97–104.
- 6 L.E. Nyquist, N.J. Hubbard, P.W. Gast, B.M. Bansal, H. Wiesmann and B. Jahn, Rb–Sr systematics for chemically defined Apollo 15 and 16 materials, *Geochim. Cosmochim. Acta, Suppl.* 4, 2 (1973) 1823–1846.
- 7 P.D. Nunes, R.J. Knight, D.M. Unruh and M. Tatsumoto, The primitive nature of the lunar crust and the problem of initial Pb isotopic compositions of lunar rocks, in: *Lunar Science V* (1974) 559–561.
- 8 N.J. Hubbard, P.W. Gast, C. Meyer, L.E. Nyquist, C. Shih and H. Weismann, Chemical composition of lunar anorthosites and their parent liquids, *Earth Planet. Sci. Lett.* 13 (1971) 71–75.
- 9 L.A. Haskin, P.A. Helmke, D.P. Blanchard, J.W. Jacobs and K. Telander, Major and trace element abundances in samples from the lunar highlands, *Geochim. Cosmochim. Acta, Suppl.* 4, 2 (1973) 1275–1296.
- 10 U. Krähenbühl, R. Ganapathy, J.W. Morgan and E. Anders, Volatile elements in Apollo 16 samples: implications for highland volcanism and accretion history of the moon, *Geochim. Cosmochim. Acta, Suppl.* 4, 2 (1973) 1325–1348.
- 11 O.B. James, Lunar anorthosite 15415: texture, mineralogy, and metamorphic history, *Science* 175 (1972) 432–436.
- 12 S. Ghose, I.S. McCallum and E. Tidy, Luna 20 pyroxenes: exsolution and phase transformation as indicators of petrologic history, *Geochim. Cosmochim. Acta* 37 (1973) 831–839.
- 13 LSPET, The Apollo 16 lunar samples: petrographic and chemical description, *Science* 179 (1973) 23–34.
- 14 G.E. Ulrich, A geologic model for North Ray Crater and stratigraphic implications for the Descartes region, *Geochim. Cosmochim. Acta, Suppl.* 4, 1 (1973) 27–39.
- 15 G. Turner, P.H. Cadogan and C.J. Yonge, Argon selenochronology, *Geochim. Cosmochim. Acta, Suppl.* 4, 2 (1973) 1889–1914.

- 16 G.M. Brown, A. Peckett, R. Phillips and C.H. Emeleus, Mineral-chemical variations in the Apollo 16 magnesianfeldspathic highland rocks. *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 505–518.
- 17 I.M. Steele and J.V. Smith, Mineralogy and petrology of some Apollo 16 rocks and fines: general petrologic model of the moon, *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 519–536.
- 18 A. El Goresy, P. Ramdohr and O. Medenbach, Lunar samples from Descartes site: opaque mineralogy and geochemistry, *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 733–750.
- 19 J.H. Scoon, Chemical analysis of lunar samples from the Apollo 16 and 17 collections, in: *Lunar Science V* (1974) 690–692.
- 20 J.L. Warner, C.H. Simonds and W.C. Phinney, Apollo 16 rocks: classification and petrogenetic model. *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 481–504.
- 21 M. Prinz, E. Dowty, K. Keil and T.E. Bunch, Mineralogy, petrology and chemistry of lithic fragments from Luna 20 fines: origin of the cumulate ANT suite and its relation to high alumina and mare basalts, *Geochim. Cosmochim. Acta* 37 (1973) 979–1006.
- 22 D. Walker, J. Longhi, T.L. Grove, E. Stolper and J.F. Hays, Experimental petrology and origin of rocks from the Descartes Highlands, *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 1013–1032.
- 23 J.V. Smith and I.M. Steele, Intergrowths in lunar and terrestrial anorthosites with implications for lunar differentiation, *Am. Miner.* 59 (1974) 673–680.
- 24 P.L. Roeder and R.F. Emslie, Olivine–liquid equilibrium, *Contr. Mineral. Petrol.* 29 (1970) 275–289.
- 25 A. Peckett, and G.M. Brown, Plutonic or metamorphic equilibration in Apollo 16 lunar pyroxenes, *Nature* 242 (1973) 252–255.
- 26 N. Morimoto and M. Tokonami, Oriented exsolution of augite in pigeonite, *Am. Miner.* 54 (1969) 1101–1117.
- 27 I.S. McCallum, F.P. Okamura and S. Ghose, Exsolution and inversion in lunar and terrestrial plutonic pyroxenes, *Trans. Am. Geophys. Union* 54 (1973) 1130.
- 28 M. Ross, P. Robinson and H.W. Jaffe, Pigeonite lamellae in Bushveld augite sequentially exsolved on phase boundaries at angles controlled by differential thermal contraction, *Geol. Soc. Am. Abstracts* 4 (1972) 644–645.
- 29 C.B. Raleigh and J.L. Talbot, Mechanical twinning in naturally and experimentally deformed diopside, *Am. J. Sci.* 265 (1967) 151–165.
- 30 I.S. McCallum, Exsolution and inversion in Stillwater pyroxenes, *Trans. Am. Geophys. Union* 55 (1974) 468.
- 31 H. Takeda, Inverted pigeonites from a clast of rock 15459 and basaltic achondrites, *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 875–885.
- 32 M. Ross, J.S. Huebner and N. Hickling, Delineation of the orthopyroxene-pigeonite transition and its bearing on pyroxene phase relations in lunar rocks, in: *Lunar Science IV* (1973) 637–639.
- 33 I. Kushiro, Determination of liquidus relations in synthetic silicate systems with electron probe analysis: the system forsterite-diopside-silica at 1 atmosphere, *Am. Miner.* 57 (1972) 1260–1271.
- 34 D. Smith, Stability of iron-rich pyroxene in the system  $\text{CaSiO}_3\text{--FeSiO}_3\text{--MgSiO}_3$ , *Am. Miner.* 57 (1972) 1413–1428.
- 35 T. Ishii and H. Takeda, Inversion, decomposition and exsolution phenomena of terrestrial and extraterrestrial pigeonites, *Mem. Geol. Soc. Japan*, 11 (1974) 19–36.
- 36 M. Ross, J.S. Huebner and E. Dowty, Delineation of the one atmosphere augite-pigeonite miscibility gap for pyroxenes from lunar basalt 12021, *Am. Miner.* 58 (1973) 619–635.
- 37 D.H. Lindsley, H.E. King and A.C. Turnock, Compositions of synthetic augite and hypersthene coexisting at 810°C: application to pyroxenes from lunar highland rocks, *Geophys. Res. Lett.* 1 (1974) 134–136.
- 38 F.N. Hodges and I. Kushiro, Petrology of Apollo 16 lunar highland rock, *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 1033–1048.
- 39 J.C. Drake, Mineralogy and chemistry of 61016,215, in: *Lunar Science V* (1974) 177–179.
- 40 R.B. Hargraves and L.S. Hollister, Mineralogic and petrologic study of lunar anorthosite slide 15415,18 *Science* 175 (1972) 430–432.
- 41 I.S. McCallum, Equilibrium relationships among coexisting minerals in the Stillwater complex, Montana, Ph. D. Thesis, Univ. of Chicago (1968) 175 pp.
- 42 L.R. Wager and G.M. Brown, Layered Igneous Rocks (Freeman, 1967) 588 pp.
- 43 G.J. Taylor, M.J. Drake, J.A. Wood and U.B. Marvin, The Luna 20 lithic fragments, and the composition and origin of the lunar highlands, *Geochim. Cosmochim. Acta* 37 (1973) 1087–1106.
- 44 A.M. Reid, J.L. Warner, W.I. Ridley and R.W. Brown, Luna 20 soil: abundance and composition of phases in the 45–125 micron fraction, *Geochim. Cosmochim. Acta* 37 (1973) 1011–1030.
- 45 P. Jakeš and A.M. Reid, Chromium partitioning between olivine and pyroxene and the redox state of lunar rocks, in: *Lunar Science V* (1974) 381–383.
- 46 S.E. Haggerty, F.R. Boyd, P.L. Bell, L.W. Finger and W.B. Bryan, Opaque minerals and olivine in lavas and breccias from Mare Tranquillitatis, *Geochim. Cosmochim. Acta*, Suppl. 1, 1 (1970) 513–538.
- 47 F.R. Boyd and D. Smith, Composition zoning in pyroxenes from lunar rock 12021, *Oceanus Proc. Oceanogr.* 12 (1971) 439–464.
- 48 D.F. Weill, R.A. Grieve, I.S. McCallum and Y. Bottinga, Mineralogy-petrology of lunar samples; microprobe studies of samples 12021 and 12022; viscosity of melts of selected lunar compositions, *Geochim. Cosmochim. Acta*, Suppl. 2, 1 (1971) 413–430.
- 49 R.G. Burns, D.J. Vaughan, R.M. Abu-Eid and M. Witner, Spectral evidence for  $\text{Cr}^{3+}$ ,  $\text{Ti}^{3+}$ , and  $\text{Fe}^{2+}$  rather than  $\text{Cr}^{2+}$  and  $\text{Fe}^{3+}$  in lunar ferromagnesian silicates, *Geochim. Cosmochim. Acta*, Suppl. 4, 1 (1973) 983–994.

- 50 R.G. Burns, *Mineralogical Applications of Crystal Field Theory* (Cambridge Univ. Press, 1970) 225 pp.
- 51 D. Walker, T.L. Grove, J. Longhi, E.M. Stolper and J.F. Hays, Origin of lunar feldspathic rocks, *Earth Planet. Sci. Lett.* 20 (1973) 325–336.
- 52 P.L. Roeder and E.F. Osborn, Experimental data for the system  $\text{MgO}-\text{FeO}-\text{Fe}_2\text{O}_3-\text{CaAl}_2\text{Si}_2\text{O}_8-\text{SiO}_2$  and their petrologic implications, *Am. J. Sci.* 264 (1966) 428–480.
- 53 E. Dowty, M. Prinz and K. Keil, Ferroan anorthosite, a widespread and distinctive lunar rock type, *Earth Planet. Sci. Lett.* 24 (1974) 15–25.
- 54 A. Stettler, P. Eberhardt, J. Geiss, N. Grögler and P. Maurer,  $^{39}\text{Ar}$ - $^{40}\text{Ar}$  ages and  $^{37}\text{Ar}$ - $^{38}\text{Ar}$  exposure ages of lunar rocks. *Geochim. Cosmochim. Acta, Suppl.* 4, 2 (1973) 1865–1888.
- 55 O.A. Schaeffer and L. Husain, Chronology of lunar basin formation and ages of lunar anorthositic rocks, in: *Lunar Science V* (1974) 663–665.
- 56 D.A. Papanastassiou and G.J. Wasserburg, Rb–Sr systematics of Lunar 20 and Apollo 16 samples, *Earth Planet. Sci. Lett.* 17 (1972) 52–63.
- 57 F. Tera, D.A. Papanastassiou and G.J. Wasserburg, A lunar cataclysm at  $\sim 3.95$  AE and the structure of the lunar crust, in: *Lunar Science IV* (1973) 723–725.
- 58 O.A. Schaeffer and L. Husain, Early lunar history: ages of 2–4 mm soil fragments from the lunar highlands, *Geochim. Cosmochim. Acta, Suppl.* 4, 2 (1973) 1847–1863.
- 59 D.A. Papanastassiou and G.J. Wasserburg, The Rb–Sr age of a crystalline rock from Apollo 16, *Earth Planet. Sci. Lett.* 16 (1972) 289–298.
- 60 C.M. Gray, D.A. Papanastassiou and G.J. Wasserburg, The identification of early condensates from the solar nebula, *Icarus* 20 (1973) 213–239.
- 61 A.L. Albee, A.A. Chodos, R.F. Dymek, A.J. Gancarz, D.S. Goldman, D.A. Papanastassiou and G.J. Wasserburg, Dunite from the lunar highlands: petrography, deformational history, Rb–Sr age, in: *Lunar Science V* (1974) 3–5.
- 62 D.F. Weill, G.A. McKay, S. Kridelbaugh and M. Grutzeck, Modeling the evolution of Sm and Eu abundances during lunar igneous differentiation, *Geochim. Cosmochim. Acta, Suppl.* 5, 2 (1974) 1337–1352.
- 63 G.B. Hess, Heat and mass transport during crystallization of the Stillwater igneous complex, *Geol. Soc. Am. Mem.* 132 (1972) 503–520.
- 64 I.S. McCallum, F.P. Okamura, E.A. Mathez and S. Ghose, Pyroxene relations in highland plutonic and high grade metamorphic rocks, in: *Lunar Science V* (1974) 472–474.